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/// Habit Tracker Application
class HabitTracker

    /// Declarations
    string userID
    string password
    string menuChoice
    string habitName
    string selectedHabit
    string category
    num completedDays
    num habitCount = 0
    num i = 0

    /// Arrays store habit information for multiple habits
    string habitList[50]
    num completedDaysList[50]
    string categoryList[50]

    string progressMessage

    /// Start
    start

    print "Welcome to Habit Tracker"

    /// Main program loop continues until user exits the app
    while menuChoice <> "Exit"

        /// Display main menu options in the GUI
        print "Main Menu"
        print "Add Habit"
        print "View Habits"
        print "Mark Habit Complete"
        print "Delete Habit"
        print "View Habit Progress"
        print "Exit"

        print "Select an option:"
        input menuChoice

    /// User creates a new habit and stores it in arrays
    if menuChoice = "Add Habit" then

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print "Enter habit name:"
input habitName

print "Enter habit category:"
input category

/// Store habit data in matching array positions
habitList[habitCount] = habitName
categoryList[habitCount] = category
completedDaysList[habitCount] = 0

/// Increase total number of saved habits
habitCount = habitCount + 1

print "Habit added to user habit list"

/// Display all saved habits and progress
else if menuChoice = "View Habits" then

    /// Decision making checks if habits exist
    if habitCount > 0 then

        print "User Habit List:"

        i = 0

        /// Iteration loop displays all habits
        while i < habitCount

            print habitList[i], " - ", completedDaysList[i], " days"

            i = i + 1

        endwhile

    else

        print "No habits in the list"

    endif

/// User marks a habit as completed for the day
else if menuChoice = "Mark Habit Complete" then

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if habitCount > 0 then

    print "Select a habit to mark complete:"

    /// Begin habit selection process
    i = 0

    /// Loop through all habits for selection
    while i < habitCount

        print habitList[i]

        i = i + 1

    endwhile

    input selectedHabit

    i = 0

    /// Search for the selected habit
    while i < habitCount

        if habitList[i] = selectedHabit then

            /// Increase completed day count
            completedDaysList[i] = completedDaysList[i] + 1

            /// Decision making based on habit progress
            if completedDaysList[i] = 21 then

                progressMessage = "Congratulations! Your habit is formed."

            else if completedDaysList[i] > 21 then

                progressMessage = "You are maintaining this habit."

            else

                progressMessage = "Keep going."

            endif

            /// Output progress message to user

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        print progressMessage

    endif

    i = i + 1

endwhile

else

    print "No habits available"

endif

/// User deletes a selected habit
else if menuChoice = "Delete Habit" then

    if habitCount > 0 then

        print "Select a habit to delete:"

        i = 0

        while i < habitCount

            print habitList[i]

            i = i + 1

        endwhile

        input selectedHabit

        i = 0

        /// Locate habit and shift remaining array values
        while i < habitCount

            if habitList[i] = selectedHabit then

                /// Move array elements to close deleted position
                while i < habitCount - 1

                    habitList[i] = habitList[i + 1]

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        categoryList[i] = categoryList[i + 1]
        completedDaysList[i] = completedDaysList[i + 1]

        i = i + 1

    endwhile

    /// Reduce total habit count
    habitCount = habitCount - 1

    print "Habit deleted"

    i = habitCount

else

    i = i + 1

endif

endwhile

else

    print "No habits to delete"

endif

/// Display detailed progress for selected habit
else if menuChoice = "View Habit Progress" then

    if habitCount > 0 then

        print "Select a habit to view progress:"

        i = 0

        while i < habitCount

            print habitList[i]

            i = i + 1

        endwhile

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input selectedHabit

i = 0

/// Search for selected habit and display progress
while i < habitCount

    if habitList[i] = selectedHabit then

        /// GUI output for habit details
        print "Habit Name: ", habitList[i]
        print "Category: ", categoryList[i]
        print "Completed Days: ", completedDaysList[i]

        /// Display motivational progress messages
        if completedDaysList[i] = 21 then

            print "Congratulations! Your habit is formed."

        else if completedDaysList[i] > 21 then

            print "You are maintaining this habit."

        else

            print "This habit is still in progress."

        endif

    endif

    i = i + 1

endwhile

else

    print "No habits available"

endif

else if menuChoice = "Exit" then

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        print "Exiting Habit Tracker"

    else

        print "Invalid option. Please try again."

    endif

endwhile

print "Thank you for using Habit Tracker"

stop

classEnd
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